

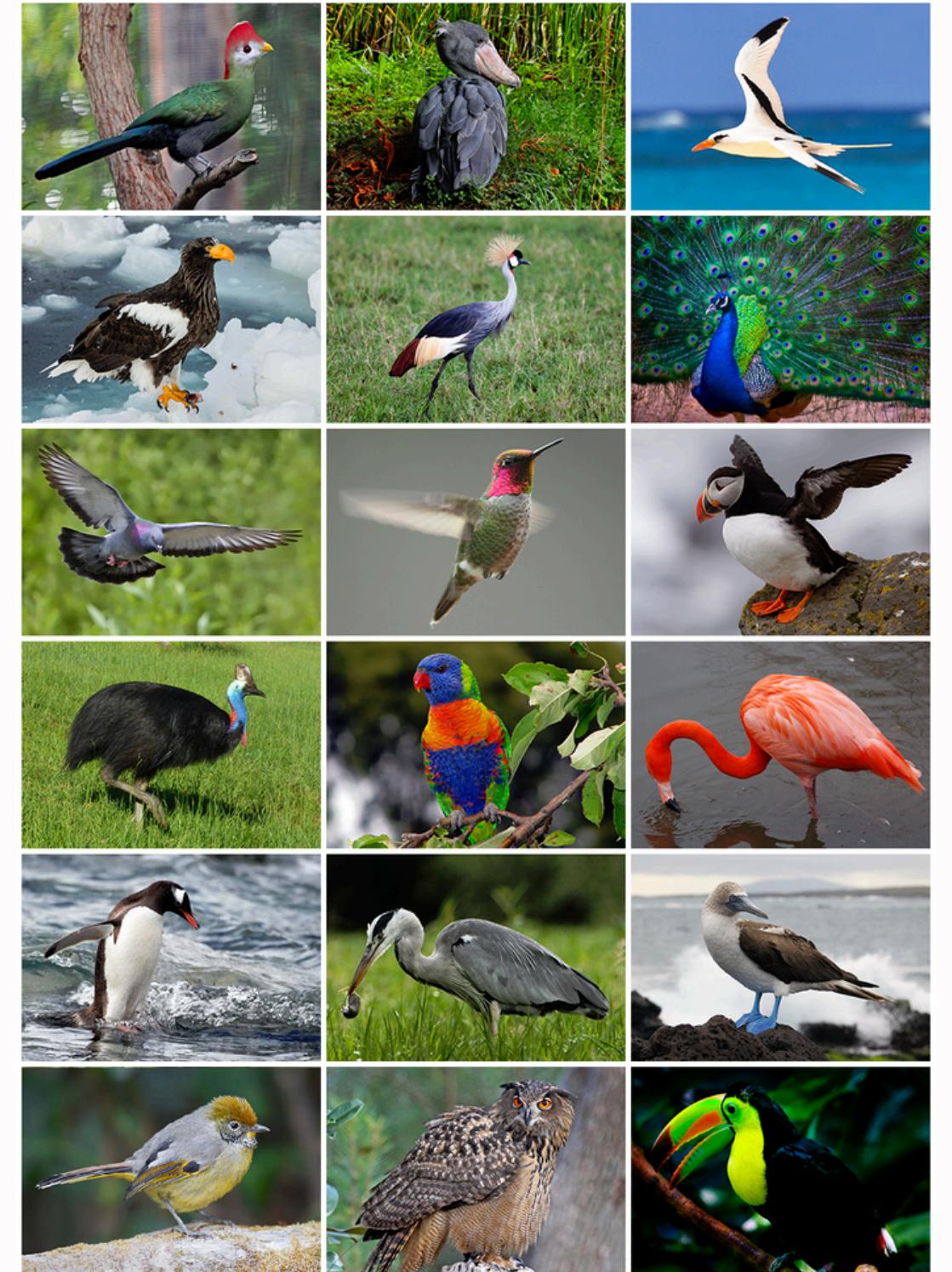


Smithsonian

Presentation 2023



UCSB-SMITHSONIAN SCHOLARS HIMALAYAN PROJECT





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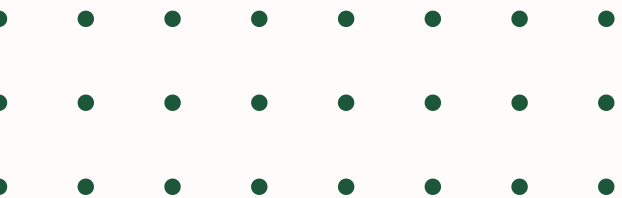
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QUESTION 1 & 2

WRITTEN RESPONSE

Review the data table's structure and describe any key elements and initial challenges your research team encountered.

a) Share your unique solutions to the challenges

b) What information is shared between each table?

c) Would you recommend any tidydata best practices for either table? How do these tables represent good tidydata practices?

a.) The unique solutions are presented in the code, which will be explained later with greater detail throughout the slides, but it was challenging to plot the graphs accurately.

b.) The information shared between each table is the grid, region, bird species and in some cases the elevation and the annual precipitation.

c.) One of the best practices of each table is the use of effective and understandable data table/frames names.

2. Explain the relationship between the data frames and how they may be linked?

Merged Data

```
69 {r}
70 merged_data <- inner_join(forest_metadata, birds ,by = c("grid" = "X1"))
71 #Explanation of Code Chunk: This code chunk merged both the forest_metadata and birds data
    tables by setting the grd and X1 equal to each other.
72 {r}
```

	grid	elevation	lat	long	region	max_temp_of_warmest_month	temp_annual_range
1	B3	153.5	26.63125	89.58324	E	311	207
2	B2	164.3	26.62938	89.58296	E	322	221
3	A2	198.3	26.97898	92.92198	E	319	219
4	B4	209.8	26.89642	88.85593	E	308	206
5	A8	300.0	26.96138	93.01216	E	320	219
6	A3	734.3	27.00627	92.40457	E	289	223

The foundation of code that allowed us to work of the rest of the project questions.

temp_annual_range	annual_precip	precip_seasonality	Macropygia_unchall	Streptopelia_chinensis	St
207	4209	106	0	0	
221	3266	111	0	0	
219	2389	86	0	0	
206	3597	111	0	0	
219	2445	85	0	0	
223	2209	90	0	0	
223	2209	90	0	0	
NA	NA	NA	0	0	
182	2603	108	0	0	
182	2603	108	0	0	
249	1357	90	0	0	
205	1754	94	4	0	
250	1361	90	0	0	

Question 3a

R Code:

```
```{r}
#3a) Bird Data Frame - grid, region, genus, n
bird_data_frame <- merged_data %>%
 select(-lat, -elevation, -long, -max_temp_of_warmest_month, -temp_annual_range,
 -annual_precip, -precip_seasonality) %>%
 pivot_longer(names_to = "genus", values_to = "n", cols = c(-grid, -region)) %>%
 filter(n > 0)
```

## Bird Grid

	grid	total_birds
row names		138
2	A3	172
3	A4	239
4	A6	238
5	A7	219
6	A8	94
7	B1	249
8	B2	112
9	B3	103
10	B4	92
11	B5	191
12	D1	86
13	D3	27
14	G1	164
15	J1	134
16	J2	116
17	J4	128

## Bird Region

	region	total_birds
1	E	2848
2	W	2120

There seems to be more birds in the East than West, about 2848 birds in the East and 2120 in the West.

```
82 ```{r}
83 #Birds Grid - Grid, Total Birds
84 birds_grid <- bird_data_frame %>%
85 group_by(grid) %>%
86 summarize(total_birds = sum(n))
87 #Explanation of code chunk: From the bird_data_frame we created the bird_grid data frame
88 and grouped the columns and rows by grid, then summarized(into one value) by total number of
89 birds using the sum function.
89 ```{r}
90 #Birds Region - Region, Total Birds - There seems to be more birds in the East than West,
91 about 2848 birds in the East and 2120 in the West.
91 birds_region <- bird_data_frame %>%
92 group_by(region) %>%
93 summarize(total_birds = sum(n))
```



# 3b Questions

b) How many bird species were present in each grid?

	region	grid	num_species
1	E	A2	34
2	E	A3	44
3	E	A4	48
4	E	A6	39
5	E	A7	35
6	E	A8	28
7	E	B1	43

```
102 bird_species_per_grid <- bird_data_frame %>%
103 group_by(region, grid) %>%
104 summarise(num_species = n_distinct(genus))
```

*How many bird species were present overall in each region?*

About 216 bird species were present in the East and 163 in the West.

	region	num_species
1	E	216
2	W	163

```
109 bird_species_per_region <- bird_data_frame %>%
110 group_by(region) %>%
111 summarise(num_species = n_distinct(genus))
```





# 3c Questions



Which grid had the highest number of bird species in each region?

Solution: In the East grid A4 has 48 bird species and demonstrates the highest number of bird species. While the West grid U1 has 45 species.

```
124 grid_with_highest_species <- bird_species_per_grid %>%
125 group_by(region) %>%
126 filter(num_species == max(num_species))
```

	region	grid	num_species
1	E	A4	48
2	W	U1	45

What is the elevation and precipitation of each of those grid locations?

Solution: The highest populated Eastern Location that has 48 bird species has an elevation of 1243.3 and an annual precipitation of 2209. While the West has 45 species, an elevation of 435 and an unknown annual precipitation.

```
133 grid_species_summary1 <- merge(grid_with_highest_species, merged_data %>%select(region,
grid, elevation, annual_precip), by = c("region", "grid"))
```

	region	grid	num_species	elevation	annual_precip
1	E	A4	48	1243.3	2209
2	W	U1	45	435.0	NA

Name the three most common bird species in each of those grid locations?

Solution: The 3 most common bird species in the highest populated Eastern grid location A4 of 48 bird species is the *Aceros\_nipalensis*, *Actinodura\_egertoni*, and the *Aethopyga\_saturat*. while in the highest populated Western grid location U1 of 45 bird species is the *Alcippe\_nipalensis*, *Caprimulgus\_macrurus*, and the *Cettia\_palli*.

	region	grid	num_species.x	elevation	annual_precip	num_species.y	most_common_species
1	E	A4	48	1243.3	2209	48	<i>Aceros_nipalensis</i> , <i>Actinodura_egertoni</i> , <i>Aethopyga_saturata</i>
2	W	U1	45	435.0	NA	45	<i>Alcippe_nipalensis</i> , <i>Caprimulgus_macrurus</i> , <i>Cettia_pallidipes</i>

```
133 grid_species_summary2 <- bird_data_frame %>%
134 group_by(region, grid) %>%
135 summarise(num_species = n_distinct(genus), most_common_species = toString(names(sort(table(genus), decreasing = TRUE)[1:3])))
136 grid_species_summary <- left_join(grid_species_summary1, grid_species_summary2, by = c("region", "grid"))
137
```

# 3d Questions

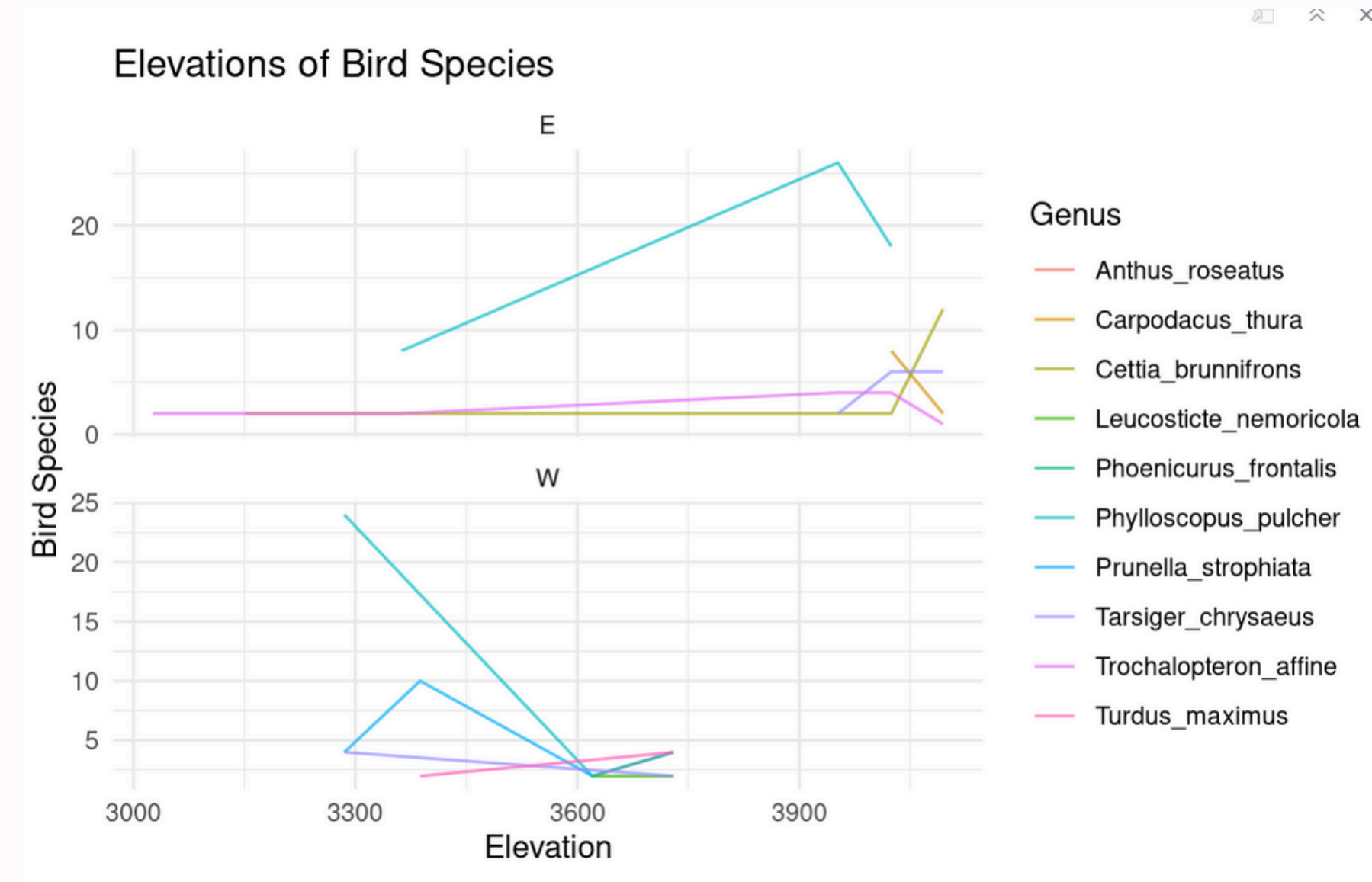
d) Within each region, which grid location had the least amount of bird species observed? Which species were observed in each of those grid locations? Visualize all the elevations where those bird species are found. Separate this by region. Explain the trends that you see on the graph (Hint: use a line graph and color by species).

The lowest populated Eastern grid location D3 has 5 bird species, while the lowest populated Western grid location M4 has 6 bird species.

```

153 grid_with_lowest_species <- bird_species_per_grid %>%
154 group_by(region) %>%
155 filter(num_species == min(num_species))
156 species_in_least_species_grid <- bird_data_frame %>%
157 filter(region %in% grid_with_lowest_species$region, grid %in%
158 grid_with_lowest_species$grid)
158 species_in_least_species_grid <- merge(species_in_least_species_grid, merged_data %>%
159 select(grid, elevation), by = "grid")
159 genus_elevation_dataframe <- merged_data %>%
160 select(elevation, region, Tarsiger_chrysaeus, Cettia_brunnifrons, Trochalopteron_affine,
161 Carpodacus_thura, Prunella_strophiata, Phoenicurus_frontalis, Turdus_maximus,
162 Phylloscopus_pulcher, Leucosticte_nemoricola, Anthus_roseatus) %>%
161 pivot_longer(names_to = "Genus", values_to = "n", cols = c(-elevation, -region)) %>%
162 filter(n > 0)
163 ggplot(genus_elevation_dataframe, aes(x = elevation, y = n, color = Genus)) +
164 geom_line(alpha = 0.7) +
165 facet_wrap(~ region, scales = "free_y", ncol = 1) +
166 labs(x = "Elevation", y = "Bird Species", title = "Elevations of Bird Species") +
167 theme_minimal()
168

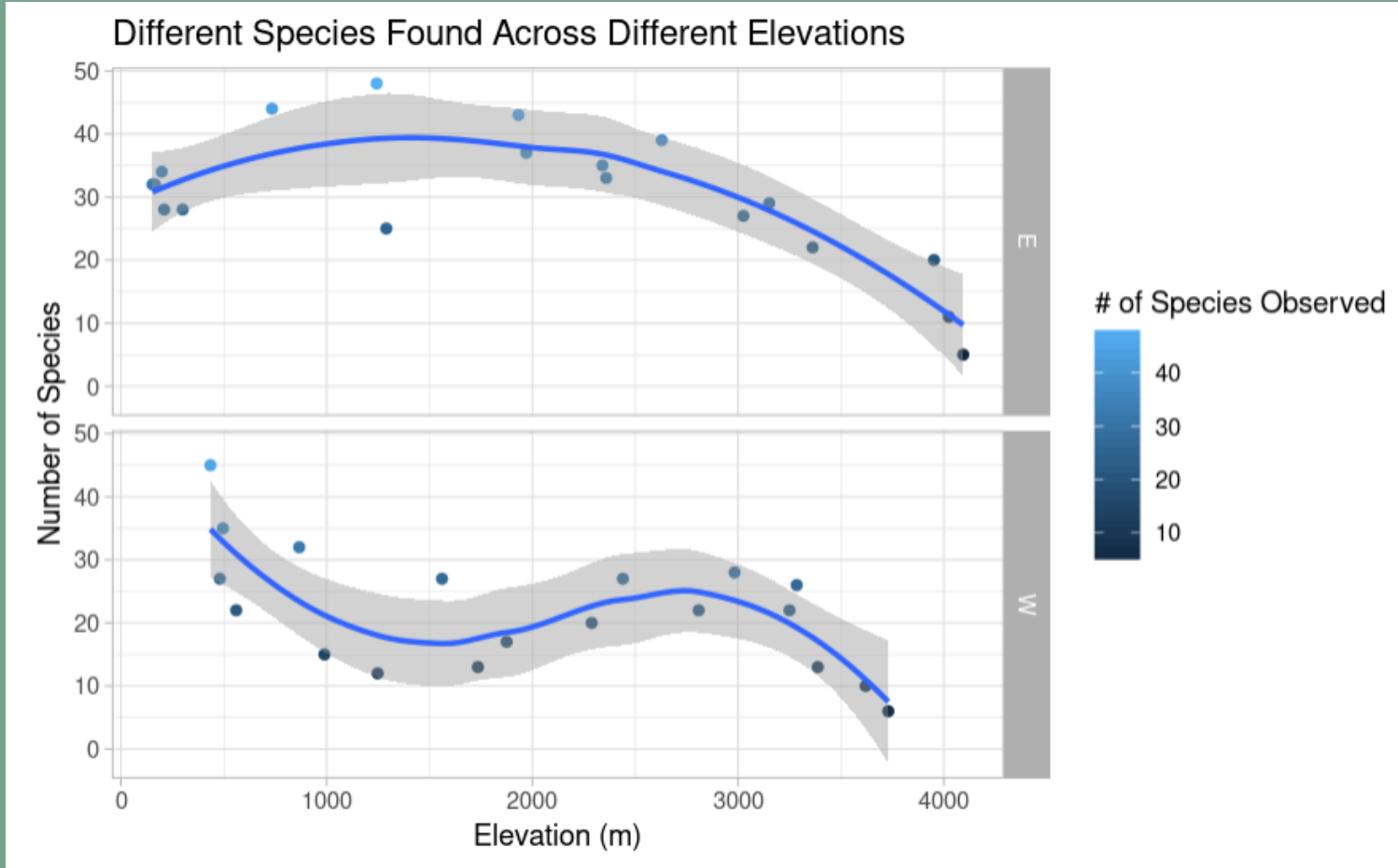
```





# Question 4a and 4b

a) When you compare and contrast your analysis for the West vs. East do you notice any trends?



## Explanation:

The Graph depicting the Number of Species observed through the different Elevation Levels when comparing the East vs. West elevations we notice that there is a trend for more species to be densely populated in the upper elevation heights in the Eastern areas compared to the Western areas, which varies greatly and has more of a diverse result in species observed through elevations.

## R Code:

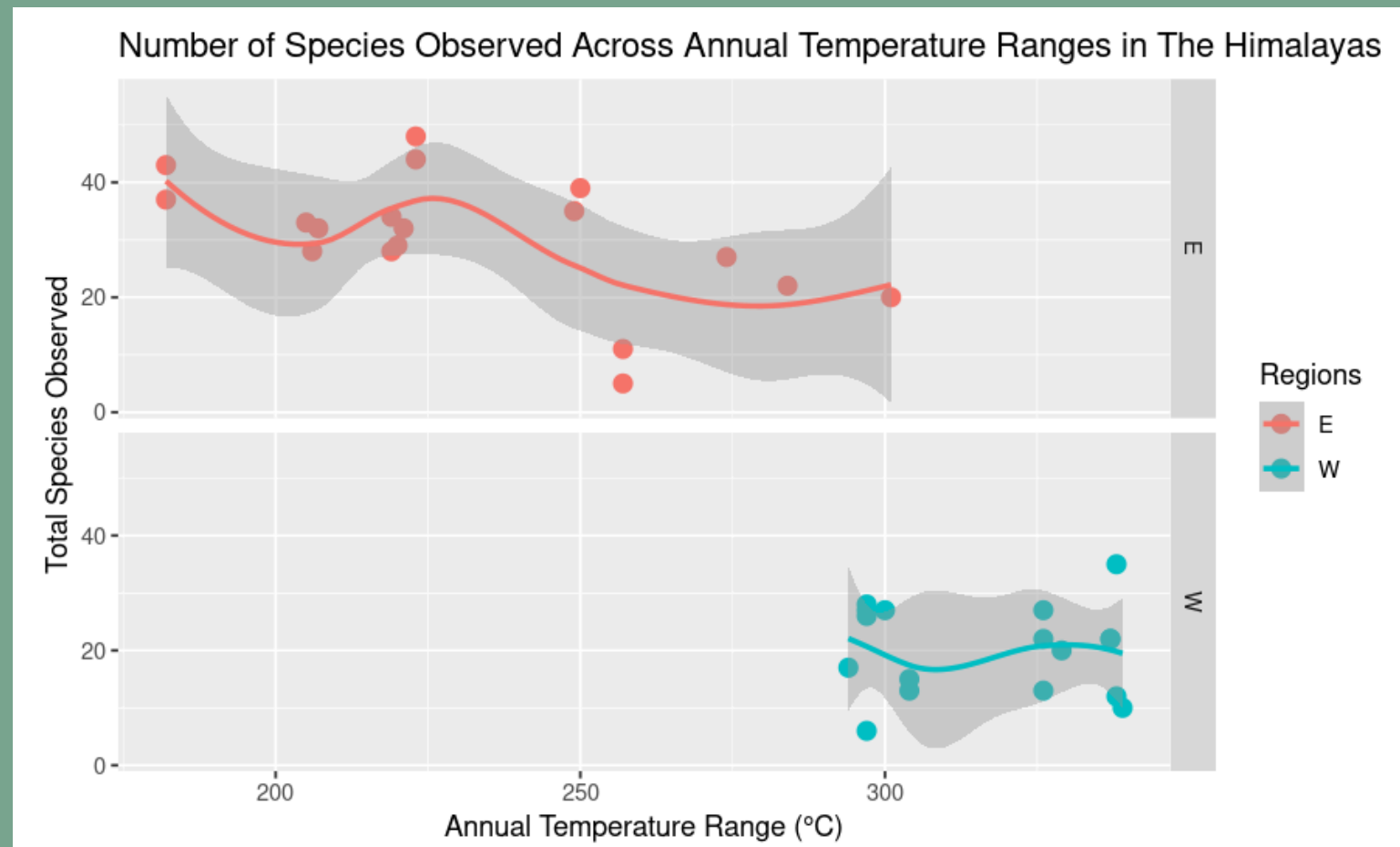
```
178 total_individual_bird_by_elevation <- merged_data %>%
179 select(-lat, -long, -max_temp_of_warmest_month, -temp_annual_range, -annual_precip,
180 -precip_seasonality) %>%
181 pivot_longer(names_to = "genus", values_to = "n", cols = c(-grid, -elevation, -region))
182 %>%
183 filter(n > 0)
184 total_individual_bird_by_elevation
185
186 # Total Number of Species Across Elevational Gradient
187 total_species_by_elevation <- total_individual_bird_by_elevation %>%
188 group_by(elevation, region, grid) %>%
189 count()
190 total_species_by_elevation
191
192 # Graph depicting the Number of Species observed through the different Elevation Levels
193 total_species_by_elevation %>%
194 ggplot(aes(x = elevation, y = n, color = n))+
195 geom_point()+
196 facet_grid(facets = vars(region))+
197 geom_smooth()+
198 theme_light()+
199 labs(x = "Elevation (m)", y = "Number of Species", color = "# of Species Observed",
200 title = "Different Species Found Across Different Elevations")
201 ```
```



# Question 5a

a) What climate data did you find most insightful to interpreting the bird data? Please explain.

R Code:



## Explanation:

We found that observing the Annual Temperature Range was most insightful to us because this Bioclimatic data displays the differences very easily between the two regions that are under question. From the video that Dr. White provided, he stated that the Western regions are much more seasonal than the Eastern region, which describes the number of bird species in those regions because of convenience for the birds dealing with temperature and climate changes. Which could also explain how some bird species are more adaptive to some conditions than others.

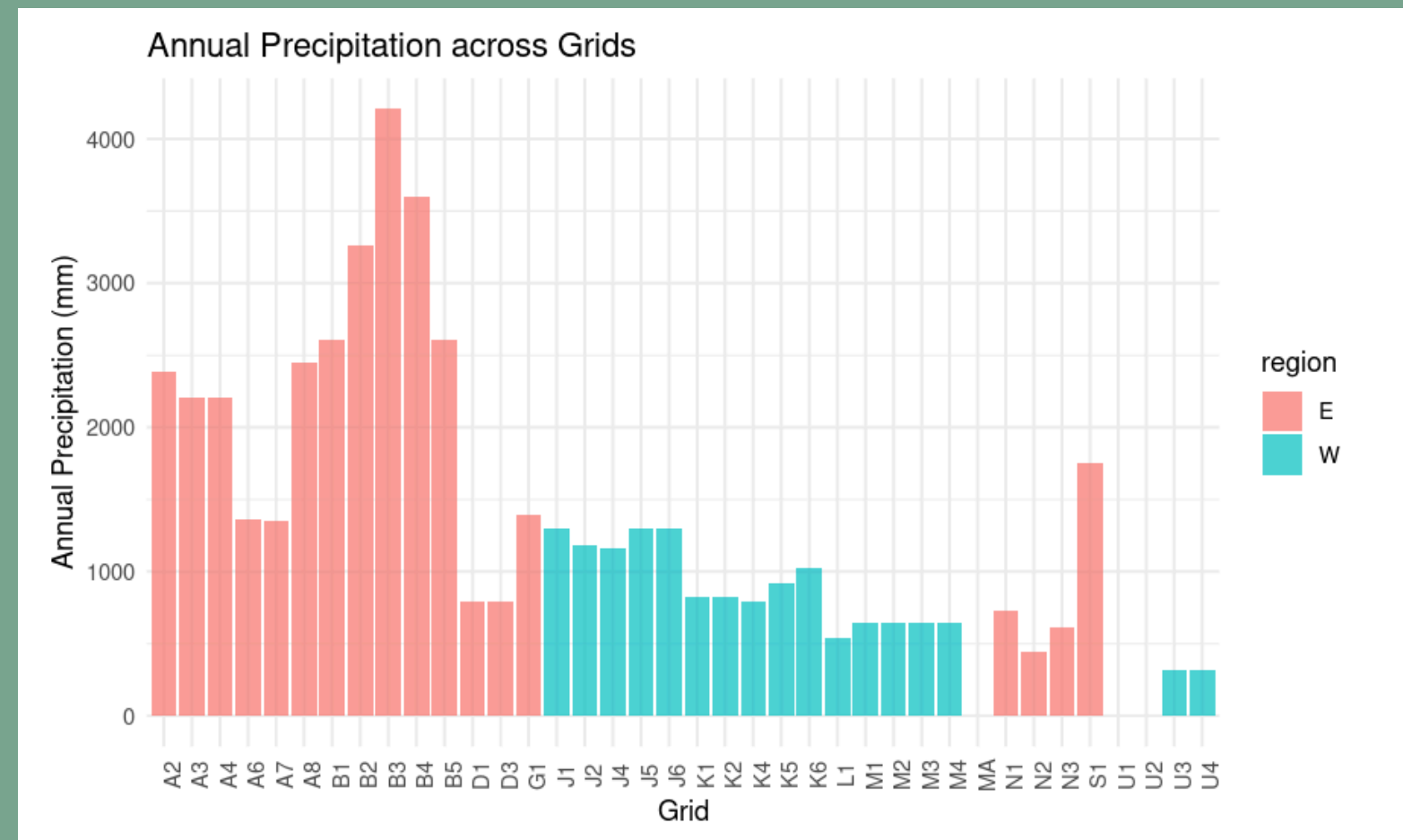
```
219 {r}
220 # Temperature with what genus and how many species counted that were observed
221 Temp_and_Bird_Species <- merged_data %>%
222 select(-lat, -elevation, -long, -annual_precip, -precip_seasonality) %>%
223 pivot_longer(names_to = "genus", values_to = "n", cols = c(-grid, -region,
224 -temp_annual_range, -max_temp_of_warmest_month)) %>%
225 filter(n > 0)
226 Temp_and_Bird_Species
227 # Max temperature of each grid and the Number of species observed in each grid
228 Temp_with_Most_Species <- Temp_and_Bird_Species %>%
229 group_by(region, grid, temp_annual_range, max_temp_of_warmest_month) %>%
230 count()
231 Temp_with_Most_Species
232 # Highest Number of Species Observed and the average temperature/max temp of the
233 region/grid they were found in
234 Temp_with_Most_Species_East_and_West <- Temp_with_Most_Species %>%
235 group_by(region) %>%
236 na.omit() %>%
237 filter(n == max(n))
238 Temp_with_Most_Species_East_and_West
239
240 Temp_with_Most_Species %>%
241 ggplot(aes(x = temp_annual_range, y = n, color = region))+
242 geom_point()+
243 facet_grid(facets = vars(region))+
244 geom_smooth()+
245 theme_gray()+
246 labs(x = "Annual Temperature Range (°C)", y = "Total Species Observed", color =
247 "Regions", title = "Number of Species Observed Across Annual Temperature Ranges in The
248 Himalayas")
```

This graph displays the overall species observed across the annual temperatures in the Himalayas.



# Question 6

Choose your own adventure! Present an additional visualization or trend in the data that you found useful in understanding the Himalayan region



## Explanation:

The graph depicts the annual precipitation across both regions. From this we can see that the East has heavier rainfall throughout the year, which may have an affect on the amount of species in the area because of food and water sourse.

## R Code:

```
250 # Annual Precipitation with what genus and how many species counted that were observed
251 Annual_Prec_and_Bird_Species <- merged_data %>%
252 select(-lat, -elevation, -long, -temp_annual_range, -max_temp_of_warmest_month,
 -precip_seasonality) %>%
253 pivot_longer(names_to = "genus", values_to = "n", cols = c(-grid, -region,
 -annual_precip)) %>%
254 filter(n > 0)
255 Annual_Prec_and_Bird_Species
256 Annual_Prec_and_Most_Bird_Species_Found <- Annual_Prec_and_Bird_Species %>%
257 group_by(region, grid, annual_precip) %>%
258 count()
259 Annual_Prec_and_Most_Bird_Species_Found
```

The following code displays the East vs. West in terms of which grid had the most species observed in comparison to how much precipitation fell in the area.

```
263 Annual_Precip_East_vs_West <- Annual_Prec_and_Most_Bird_Species_Found %>%
264 group_by(region) %>%
265 na.omit() %>%
266 filter(n == max(n))
267 ggplot(merged_data, aes(x = grid, y = annual_precip, fill = region)) +
268 geom_bar(stat = "identity", position = "dodge", alpha = 0.7) +
269 labs(x = "Grid", y = "Annual Precipitation", title = "Annual Precipitation across
 Grids") +
270 theme_minimal() +
271 theme(axis.text.x = element_text(angle = 90))
272 ```
```

**Share what next steps you might take to investigate this data further.**

We could expand the area or try our new places with a greater number of birds and see how that affects our data.



**THANK  
YOU**

